MATH 111 Calculus 1

The Mean Value Theorem

and the shape of a graph

Week 6 · Lecture 16 · §3.2, §3.3

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

STAND UP

"If f is differentiable and increasing on (a, b), then $f'(x) > 0$ for every x in (a, b)."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE. $f(x) = x^3$ is increasing on $(-1, 1)$ — if $x_1 < x_2$, then $x_1^3 < x_2^3$ — but $f'(0) = 0$. What is true: an increasing differentiable f has $f'(x) \geq 0$ (every difference quotient is positive). Today we prove the other direction: $f' > 0$ on an interval makes f increasing there (frame 12).

Rolle's Theorem

Rolle's Theorem Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$.

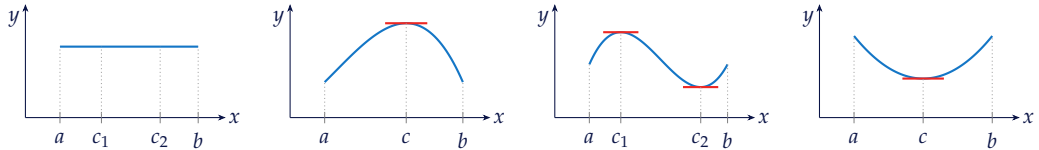
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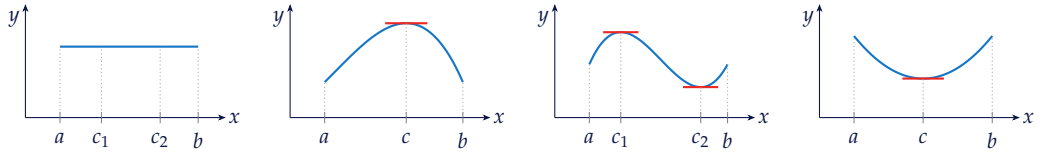


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In each picture there is at least one point $\big(c, f(c)\big)$ where the tangent is horizontal: $f'(c) = 0$. The theorem: there always is.

Rolle's Theorem — the proof

The Extreme Value Theorem: if f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value and an absolute minimum value on $[a, b]$ (taken without proof).

Fermat's Theorem: if f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.

PROOF There are three cases.

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Case I: $f(x) = k$, a constant. Then $f'(x) = 0$, so c can be *any* number in (a, b).

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Case I: $f(x) = k$, a constant. Then $f'(x) = 0$, so c can be *any* number in (a, b).

Case II: $f(x) > f(a)$ for some x in (a, b) (pictures (b), (c)). By the Extreme Value Theorem (hypothesis 1), f has a maximum value in $[a, b]$. It is larger than $f(a) = f(b)$, so it is attained at a number c in the *open* interval (a, b): a local maximum at c (an open interval around c lies in (a, b)). By hypothesis 2, $f'(c)$ exists; so $f'(c) = 0$ by Fermat's Theorem.

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Case III: $f(x) < f(a)$ for some x in (a, b) (pictures (c), (d)). The same with the minimum value: attained at a number c in (a, b), and $f'(c) = 0$ by Fermat's Theorem. ■

Using Rolle's Theorem: exactly one solution

EXAMPLE Show that the equation $x^5 + 3x - 1 = 0$ has exactly one real solution.

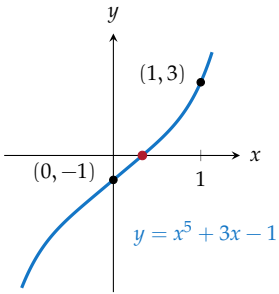
PROOF Let $f(x) = x^5 + 3x - 1$: a polynomial, so continuous and differentiable everywhere.

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At least one. $f(0) = -1 < 0$ and $f(1) = 3 > 0$. By the Intermediate Value Theorem (taken without proof), $f(c) = 0$ for some c in $(0, 1)$.



one crossing, near 0.332

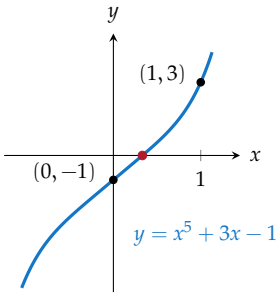
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Using Rolle's Theorem: exactly one solution

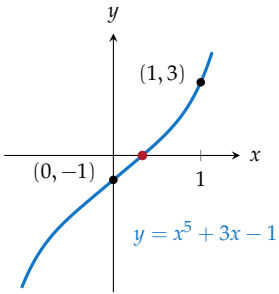
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But $f'(x) = 5x^4 + 3 \geq 3$ for every x: f' is never 0. A contradiction. Exactly one real solution. ■



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The Mean Value Theorem Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.

2. f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

(1) $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ or, equivalently, **(2)** $f(b) - f(a) = f'(c)\,(b - a)$.

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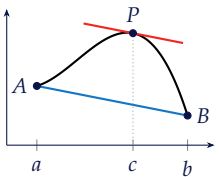
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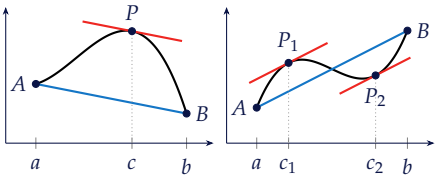
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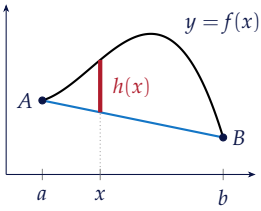
There may be several such c (right).



The Mean Value Theorem — the proof

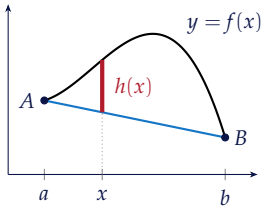
PROOF We apply Rolle's Theorem to a new function h: f minus the function whose graph is the secant line AB, $y = f(a) + \dfrac{f(b)-f(a)}{b-a}(x-a)$:

$$h(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a).$$



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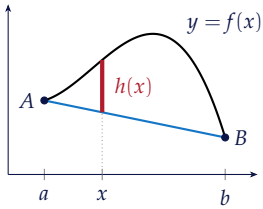


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1. h is continuous on $[a,b]$: f minus a polynomial of degree at most one.
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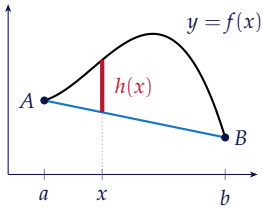


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By Rolle's Theorem, $0 = h'(c) = f'(c) - \dfrac{f(b)-f(a)}{b-a}$ for some c in (a,b):

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Question 42

Let $f(x) = x^3$ on the interval $[-1, 2]$. Which numbers c satisfy the conclusion of the Mean Value Theorem for f on $[-1, 2]$?

A. $c = \frac{1}{2}$ only

B. $c = \sqrt{3}$ only

C. $c = -1$ and $c = 1$

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In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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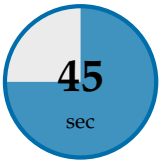
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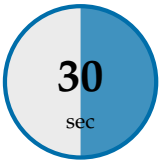
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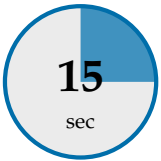
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Hands down. Thank you.



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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

D. The secant slope: $\dfrac{f(2) - f(-1)}{2 - (-1)} = \dfrac{8 - (-1)}{3} = 3$. $f'(c) = 3c^2 = 3$ gives $c = \pm 1$, and c must lie in the open interval $(-1, 2)$: $c = 1$. C keeps $c = -1$, the endpoint a, which is not in (a, b). B forgets to divide by $b - a$: $3c^2 = 9$. A takes the midpoint — "mean value" is not the midpoint: $f'(\frac{1}{2}) = \frac{3}{4} \neq 3$.

When $f' = 0$: constant

Theorem If $f'(x) = 0$ for all x in an interval (a, b), then f is constant on (a, b).

PROOF Let $x_1 < x_2$ be any two numbers in (a, b). f is differentiable on (a, b), so it is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2); the Mean Value Theorem gives a c in (x_1, x_2) with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$. So $f(x_2) = f(x_1)$. ∎

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Corollary If $f'(x) = g'(x)$ for all x in an interval (a, b), then $f - g$ is constant on (a, b); that is, $f(x) = g(x) + C$, C a constant.

PROOF $F = f - g$ has $F' = f' - g' = 0$ on (a, b); by the Theorem, F is constant. ■

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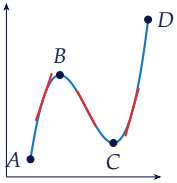
Corollary If $f'(x) = g'(x)$ for all x in an interval (a, b), then $f - g$ is constant on (a, b); that is, $f(x) = g(x) + C$, C a constant.

PROOF $F = f - g$ has $F' = f' - g' = 0$ on (a, b); by the Theorem, F is constant. ■

Caution. The interval matters: $f(x) = 1$ for $x < 0$, $f(x) = 3$ for $x > 0$ has $f'(x) = 0$ at every number of its domain, but it is not constant — its domain, all $x \neq 0$, is not an interval.

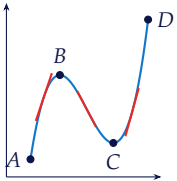
The Increasing/Decreasing Test

$f'(x)$ is the slope of the tangent at $\big(x, f(x)\big)$. From A to B and from C to D the tangents slope upwards and f increases; from B to C, downwards.



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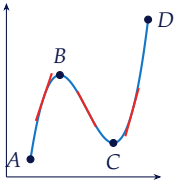


Increasing/Decreasing Test

(a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.
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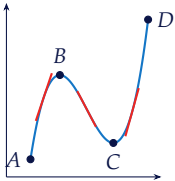
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$f' > 0$ on an interval makes f increasing — not the other way round: x^3 (frame 2).

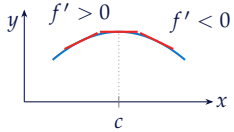
The First Derivative Test

The First Derivative Test Suppose that c is a critical point of a continuous function f.
(a) If f' changes from positive to negative at c, then f has a local maximum at c.
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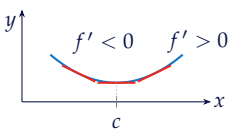
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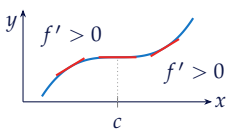
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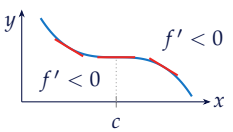
(a) local maximum



(b) local minimum



(c) neither

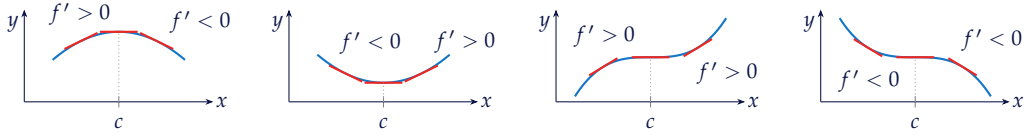


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(a) local maximum (b) local minimum (c) neither (d) neither

PROOF (a) $x < c$ near c: the Mean Value Theorem on $[x, c]$ gives $f(c) - f(x) = f'(d)(c - x) > 0$. $x > c$ near c: $f(x) - f(c) = f'(d)(x - c) < 0$. (b), (c) alike. ■

EXAMPLE $f(x) = x^4 - 8x^3 + 18x^2$. Where is f increasing or decreasing? What are its local maximum and minimum values?

$f'(x) = 4x^3 - 24x^2 + 36x = 4x\left(x^2 - 6x + 9\right) = 4x(x-3)^2$. Critical points: 0 and 3.

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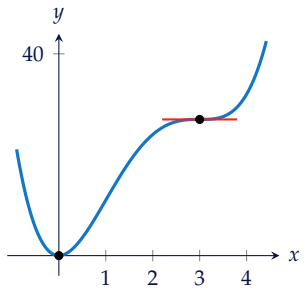
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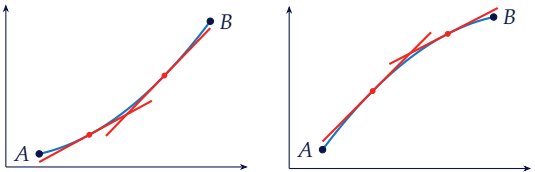
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minimum at $(0, 0)$; at $(3, 27)$ a horizontal tangent, and the graph keeps rising

What f'' says about f: concavity

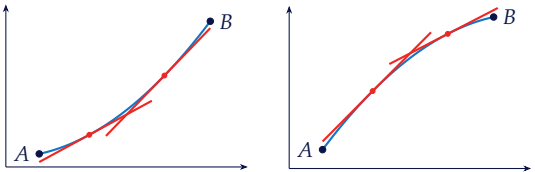
Two increasing functions from A to B, bending in different directions. Draw tangents: in (a) the curve lies above its tangents, in (b) below.



(a) concave upward (b) concave downward

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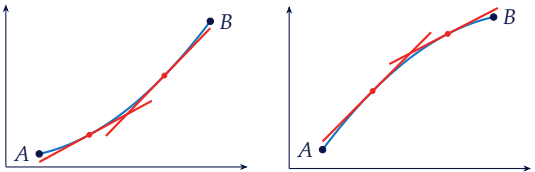


(a) concave upward (b) concave downward

Definition If the graph of f lies above all of its tangents on an interval I, then f is called **concave upward** on I. If the graph of f lies below all of its tangents on I, then f is called **concave downward** on I.

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In (a) the slope of the tangent increases from left to right: f' is increasing (here $f'' = 1.2 > 0$) — what we expect when $f'' > 0$. In (b) it decreases ($f'' = -1.2$). The next frame goes the reliable way: from the sign of f'' to the shape.

The Concavity Test; inflection points

Concavity Test

(a) If $f''(x) > 0$ on an interval I, then the graph of f is concave upward on I.
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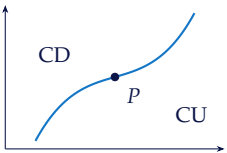
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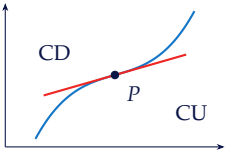
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So, by the Concavity Test: where f is continuous and f'' changes sign, there is an inflection point. If the curve has a tangent there, it crosses it (taken without proof).

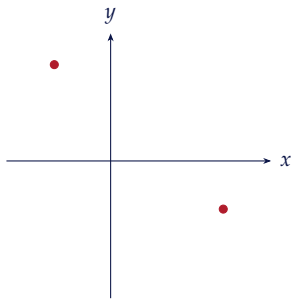
Sketching a graph from f' and f''

EXAMPLE Sketch a possible graph of a function f that satisfies the following conditions:

(i) $f(-2) = 4, f(1) = 1, f(4) = -2, f'(-2) = f'(4) = 0$

(ii) $f'(x) > 0$ for $x < -2$ and for $x > 4$,
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(iii) $f''(x) < 0$ for $x < 1$, $f''(x) > 0$ for $x > 1$

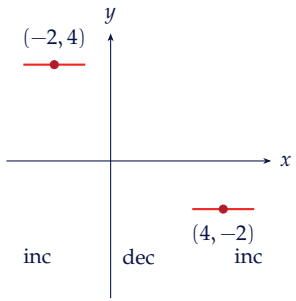


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(i) Horizontal tangents at $(-2, 4)$ and $(4, -2)$. **(ii)** f increases on $(-\infty, -2)$ and $(4, \infty)$, decreases on $(-2, 4)$ (I/D Test): local maximum $f(-2) = 4$, local minimum $f(4) = -2$ (First Derivative Test).

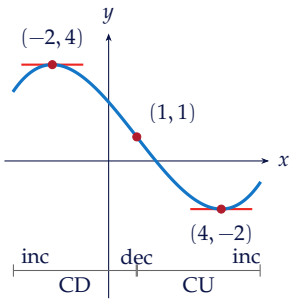
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(iii) Concave downward on $(-\infty, 1)$, upward on $(1, \infty)$ (Concavity Test): $(1, 1)$ is an inflection point.

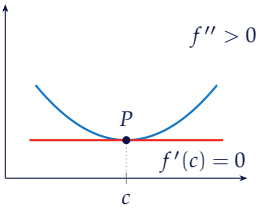
The Second Derivative Test

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(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

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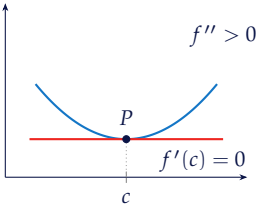
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PROOF (a) f'' is continuous at c and $f''(c) > 0$, so $f'' > 0$ on an open interval around c (continuity, with $\varepsilon = \frac{1}{2}f''(c)$). There f is concave upward (the Concavity Test): the graph lies above its tangent at c — a horizontal tangent, since $f'(c) = 0$. So $f(x) \geq f(c)$ for x near c: a local minimum.

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(b): the same with $f'' < 0$: the graph lies below its horizontal tangent. ■
And when $f''(c) = 0$? Question 43.

Question 43

f has a continuous second derivative, $f'(5) = 0$ and $f''(5) = 0$. What can you conclude about f at 5?

A. f has a local minimum at 5.

B. Any of the three can happen: a local maximum, a local minimum, or neither.

C. The point $(5, f(5))$ is an inflection point.

D. f has neither a local maximum nor a local minimum at 5.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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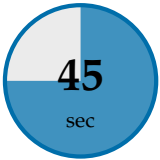
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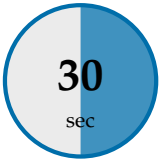
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D. f has neither a local maximum nor a local minimum at 5.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

f has a continuous second derivative, $f'(5) = 0$ and $f''(5) = 0$. What can you conclude about f at 5?

A. f has a local minimum at 5.

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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D. f has neither a local maximum nor a local minimum at 5.

Hands down. Thank you.

A B C D

Question 43

EXPLAIN

f has a continuous second derivative, $f'(5) = 0$ and $f''(5) = 0$. What can you conclude about f at 5?

A. f has a local minimum at 5.

B. Any of the three can happen: a local maximum, a local minimum, or neither.

C. The point $\big(5, f(5)\big)$ is an inflection point.

D. f has neither a local maximum nor a local minimum at 5.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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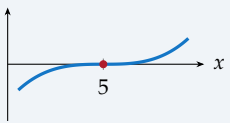
D. f has neither a local maximum nor a local minimum at 5.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

B. $(x-5)^4$, $-(x-5)^4$ and $(x-5)^3$ all have $f'(5) = 0$ and $f''(5) = 0$: the first has a local minimum at 5, the second a local maximum, the third neither. So A and D can fail; C fails for $(x-5)^4$, whose $f'' = 12(x-5)^2$ does not change sign. When $f''(c) = 0$ the Second Derivative Test gives no information: use the First Derivative Test.



$y = (x-5)^3$: neither

Question 44

f has a continuous second derivative. True or false?

If $f''(c) = 0$, then the point $\big(c, f(c)\big)$ is an inflection point of the graph of f.

In your team

Decide your own answer first — true or false — then talk. Thumb up = true, thumb down = false. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

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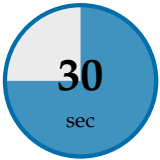
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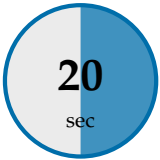
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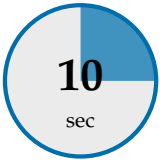
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VOTE

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Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE FALSE

VOTE

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TRUE

FALSE

VOTE

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TRUE

FALSE

VOTE

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Hands down. Thank you.

TRUE FALSE

Question 44

EXPLAIN

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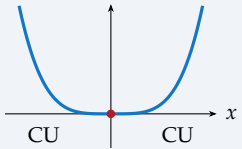
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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

False. Take $f(x) = x^4$: $f''(x) = 12x^2$, so $f''(0) = 0$. But $f''(x) > 0$ on both sides of 0: the graph is concave upward on both sides, and $(0, 0)$ is not an inflection point. An inflection point needs f'' to change sign.

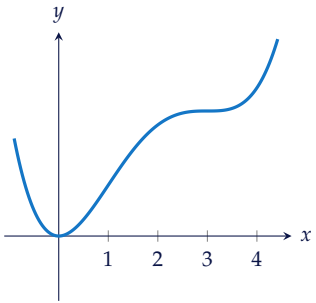


$y = x^4$: $f''(0) = 0$, no inflection

The same function, with f''

$f(x) = x^4 - 8x^3 + 18x^2$, $f'(x) = 4x(x-3)^2$,
$f''(x) = 12x^2 - 48x + 36 = 12\big(x^2 - 4x + 3\big) = 12(x-1)(x-3)$.

interval	$f''(x)$	concavity
$x < 1$	$+$	upward on $(-\infty, 1)$
$1 < x < 3$	$-$	downward on $(1, 3)$
$x > 3$	$+$	upward on $(3, \infty)$

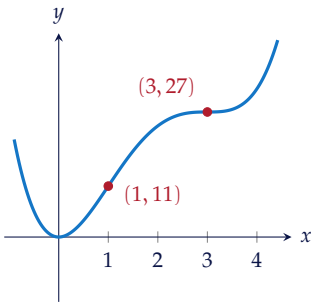


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f'' changes sign at 1 and at 3: inflection points $(1, 11)$, $(3, 27)$.

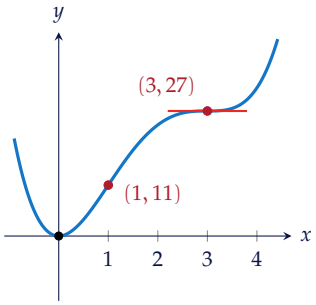


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The Second Derivative Test: $f''(0) = 36 > 0$: a local minimum at 0, as before. $f''(3) = 0$: no information (Question 43); the First Derivative Test (frame 14): no maximum or minimum at 3.

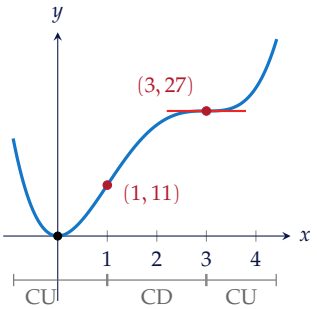
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$f(x) - 27 = (x-3)^3(x+1)$ changes sign at 3: the curve crosses its horizontal tangent — an inflection point, not an extreme value

A little history · 1690–1691

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Rolle distrusted the new calculus: he described it as a collection of ingenious fallacies, and later conceded that he was wrong. The theorem received his name only in the nineteenth century.

Today in three lines

1. For f continuous on $[a, b]$ and differentiable on (a, b): Rolle — $f(a) = f(b) \Rightarrow f'(c) = 0$ for some c in (a, b); the Mean Value Theorem — $f(b) - f(a) = f'(c)\,(b - a)$: a tangent parallel to the secant.

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This week:

Homework: §3.2 and §3.3 — read and work out the examples. Next: curve sketching (§3.5).

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"If $f' > 0$ on an interval, then f is increasing there" is true because … (one sentence — name the theorem).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.